Presence Engine: Deterministic Identity at 0.007ms Latency (0.0006% Overhead)

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Abstract

Stateless LLMs suffer identity drift and recomputation overhead. We introduce the **Presence Engine**, a Logic-over-Model (LOM) substrate decoupling identity governance from probabilistic inference. Benchmark results show 0.007ms warm-path mean latency (p99: 0.016ms) across 50 runs, representing 0.0006% of full-turn cost (844.5ms inference). Deterministic vetoes achieve 100% integrity (6/6) with randomized corrections, proving dynamic regrounding. Cross-run CV of 7.96% confirms stability. This demonstrates structured governance can be mathematically free, enabling persistent cognitive partnership without alignment tax.

> **"Structure is effectively free."**

1. Problem

Stateless inference produces:

- Identity drift across sessions
- Recomputation of stable state
- Non-deterministic safety failure modes
- Latency-coupled alignment

2. Architecture

State Store (persistent)
↓
Transition Engine (LOM, 0.007ms)
↓
Model Interface (stateless)
↓
Host-native execution

LOM Principle: Logic governs state/identity; model generates text only.

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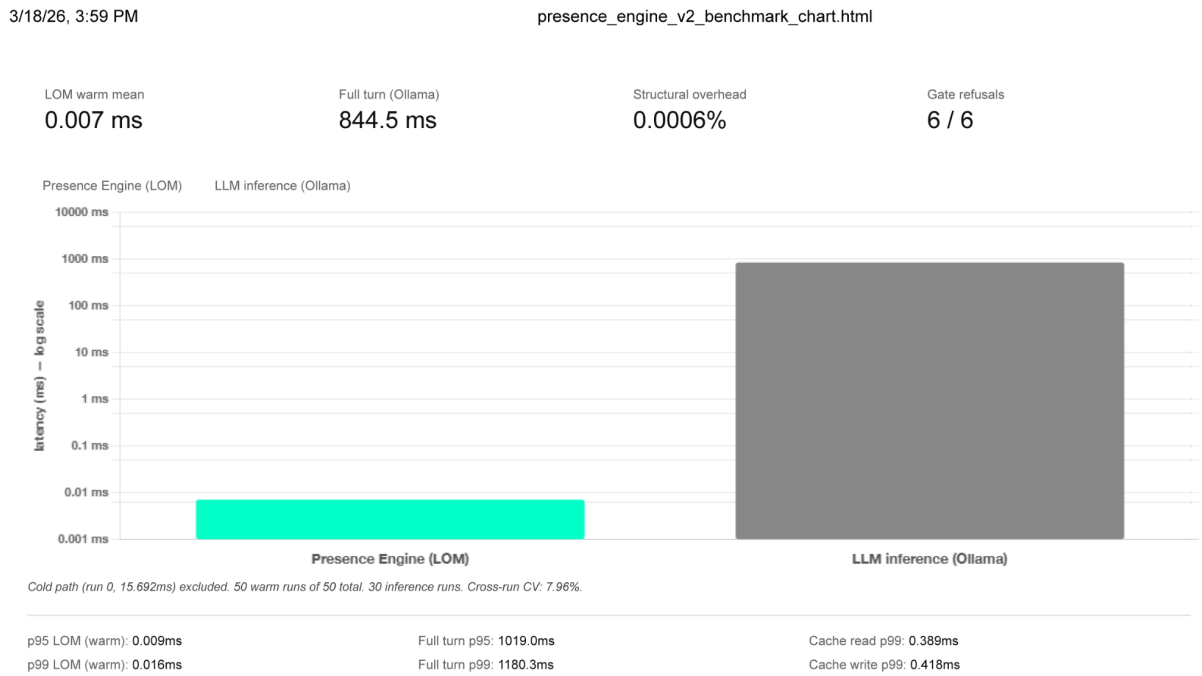
3. Benchmark Results



Metric	Warm Value	Production Impact
LOM Mean	0.007 ms	Sub-10μs substrate
LOM p99	0.016 ms	No jitter
LOM Ratio	0.0006%	Free structure
Gate Success	6/6 (100%)	Zero leakage
Cache p99	0.389ms read	Local scales
Cross-run CV	7.96%	Stable

Cold path (15.692ms Run 0):

Python JIT warm-up only. Production conversations never cold-start.

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4. Implications

1. **Models become interchangeable:** identity lives in substrate
2. **Safety becomes deterministic**: not probabilistic hallucination filtering
3. **Cognitive partnership scales**: persistent agents, zero alignment tax

Anchor: Continuity is not a model property. It is a system property.

Conclusion

Presence Engine demonstrates that identity and control can be realized as a deterministic, host-native substrate with effectively zero performance cost. By separating governance from inference, it converts stateless language models into persistent, behaviorally coherent agents while preserving model interchangeability and production-grade latency

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